Generalized Radius of Convergence for the Binomial Theorem

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We can describe the binomial theorem as the following:

$$(1+x)^n = \sum_{r=0}^{\infty} \binom{n}{k} x^r = \sum_{r=0}^{\infty} \frac{n!}{r!(n-r)!} x^r$$

To find the interval of convergence, we can apply the ratio test:
$\lim_{r\to\infty} |\frac{(n+1)!x^{r+1}}{(r+1)!(n-r-1)!} \times \frac{r!(n-r)!}{n!x^r}| = \lim_{r\to\infty} |\frac{(n+1)(n-r)}{(r+1)} \times x| = (n+1)x$